Heat Flow, Geometry, and Ricci Flow

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April 19, 2026

Overview of Presentation

- Heat flow acts as a smoothing process.
- The behavior of heat depends on the underlying space.
- Geometry is encoded by a metric.
- Curvature measures how that geometry bends.
- Ricci flow: evolve the metric the same way heat evolves temperature.

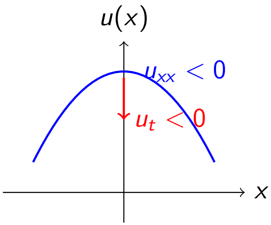
1D Heat Flow and the Laplacian

Before looking at surfaces, consider the 1D heat equation:

$$\frac{\partial u}{\partial t} = u_{xx}.$$

Why the second derivative? It measures *concavity*.

- If $u_{xx} < 0$ (concave down), temperature is a local maximum $\Rightarrow$ heat flows away ($u_t < 0$).
- If $u_{xx} > 0$ (concave up), temperature is a local minimum $\Rightarrow$ heat flows in ($u_t > 0$).



Heat Equation in $\mathbb{R}^2$

We expand this to the classical heat equation in Cartesian coordinates:

$$\frac{\partial u}{\partial t} = u_{xx} + u_{yy} = \Delta u.$$

- $u(x, y, t) =$ temperature
- Heat flows from high to low
- The Laplacian Δu measures the "average concavity" in all directions.

Boundary Conditions

To solve the heat equation on a bounded domain, we must specify what happens at the boundary ∂M.

Dirichlet Boundary Condition

Fixes the temperature at the boundary.

$$u(x, t) = 0 \quad \text{for } x \in \partial M.$$

Physical meaning: The boundary is held at a constant temperature (e.g., submerged in ice water).

Neumann Boundary Condition

Fixes the heat flux across the boundary.

$$\frac{\partial u}{\partial n}(x, t) = 0 \quad \text{for } x \in \partial M.$$

Physical meaning: The boundary is perfectly insulated; no heat enters or escapes.

Heat Equation in Polar Coordinates

In polar coordinates (r, θ), the Laplacian changes form. The heat equation becomes:

$$\frac{\partial u}{\partial t} = \Delta_{\text{polar}} u = u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta}$$

- Same physics, different coordinates
- Extra terms come from geometry
- Laplacian depends on how we measure space

Remark

This comes from rewriting the Laplacian using the chain rule. The extra terms ($\frac{1}{r} u_r$, etc.) reflect how distances and areas change as you move away from the origin.

What is a Metric?

- A metric tells us how to measure **distance, angle, and area**
- It encodes the geometry of the space

Examples

Flat (Cartesian):

$$ds^2 = dx^2 + dy^2.$$

Polar:

$$ds^2 = dr^2 + r^2 d\theta^2.$$

- Even flat space looks different under different coordinates.
- The metric determines how derivatives behave.

Laplacian Depends on Geometry

- In general, we write the Laplacian as Δ_g.
- The metric g determines how we compute derivatives.
- Different metrics $\Rightarrow$ different Laplacians.
- Therefore: geometry directly affects heat flow.

$$\text{metric} \rightarrow \text{Laplacian} \rightarrow \text{heat equation.}$$

Conformal Metrics and Weight Functions

Definition

A metric g is *conformal* to the flat metric if there exists a strictly positive weight function $u(x, y)$ such that:

$$g = u(x, y)(dx^2 + dy^2).$$

- In 1D, this looks like $g = u(x)dx^2$.
- The metric is obtained by scaling the flat metric.
- Angles are preserved, but lengths are rescaled by $\sqrt{u}$.
- Large u stretches distances; small u shrinks them.
- Geometry is encoded entirely in the weight function u.

The Graph of u vs. The Surface Itself

It is crucial to distinguish between the function $u(x, y)$ and the geometry it creates.

The Graph of u:

- The set of points $(x, y, u(x, y))$ in $\mathbb{R}^3$.
- This is a 2D surface sitting *extrinsically* inside 3D space.

The Surface defined by the metric $g = u(x, y)(dx^2 + dy^2)$:

- This defines the *intrinsic* geometry of a 2D space.
- You do not need a 3D ambient space to measure distances here.
- Walking on the xy-plane with a "stretchy ruler" defined by $u(x, y)$ is fundamentally different from walking on the 3D graph of u.

Fubini–Study Metric

A classic example of a conformal metric on the sphere $S^2 \cong \mathbb{C} \cup \{\infty\}$ is the Fubini-Study metric:

$$g_{FS} = \frac{4}{(1 + x^2 + y^2)^2}(dx^2 + dy^2).$$

Here, our conformal weight function is:

$$u(x, y) = \frac{4}{(1 + x^2 + y^2)^2}.$$

This maps the flat plane to the standard round sphere via stereographic projection.

Curvature of the Fubini-Study Metric

For a conformal metric $g = u(x, y)(dx^2 + dy^2)$, the Gaussian curvature K is given by:

$$K = -\frac{1}{2u}\Delta(\ln u).$$

Let's calculate K for Fubini-Study, where $u = 4(1 + r^2)^{-2}$ and $r^2 = x^2 + y^2$:

$$\begin{aligned}
\ln u &= \ln 4 - 2\ln(1 + r^2) \\
\Delta(\ln u) &= \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)(\ln 4 - 2\ln(1 + r^2)) \\
\partial_x \ln(1 + r^2) &= \frac{2x}{1 + r^2} \implies \partial_{xx} \ln(1 + r^2) = \frac{2(1 + r^2) - 4x^2}{(1 + r^2)^2} = \frac{2 - 2x^2 + 2y^2}{(1 + r^2)^2} \\
\Delta \ln(1 + r^2) &= \frac{2 - 2x^2 + 2y^2 + 2 - 2y^2 + 2x^2}{(1 + r^2)^2} = \frac{4}{(1 + r^2)^2} \\
\Delta(\ln u) &= -2\left(\frac{4}{(1 + r^2)^2}\right) = -\frac{8}{(1 + r^2)^2}.
\end{aligned}$$

Curvature of the Fubini-Study Metric (Cont.)

Plugging the Laplacian back into our curvature formula:

$$\begin{aligned} K &= -\frac{1}{2u}\Delta(\ln u) \\ &= -\frac{1}{2\left(\frac{4}{(1+r^2)^2}\right)}\left(-\frac{8}{(1+r^2)^2}\right) \\ &= \frac{(1+r^2)^2}{8}\cdot\frac{8}{(1+r^2)^2} \\ &= 1. \end{aligned}$$

The Fubini-Study metric describes a surface of **constant positive curvature** $K = 1$, which perfectly matches our geometric intuition of a standard sphere.

Connections and the Geodesic Equation

How do things move straight in a curved space? We need a **connection** (∇) to tell us how vectors change as we move.
The metric determines the Levi-Civita connection, expressed via Christoffel symbols Γ^k_{ij}.

The Geodesic Equation

A path $\gamma(t)$ is a geodesic (a "straight line" in curved space) if its acceleration is zero with respect to the connection:

$$\nabla_{\dot{\gamma}}\dot{\gamma} = 0.$$

In local coordinates, this gives a system of 2nd-order ODEs:

$$\frac{d^2\gamma^k}{dt^2} + \sum_{i,j} \Gamma^k_{ij} \frac{d\gamma^i}{dt} \frac{d\gamma^j}{dt} = 0.$$

The Ricci Flow Equation

Introduced by Richard Hamilton (1982), Ricci flow evolves the metric $g(t)$ to smooth out irregular curvature, governed by the Ricci curvature tensor Ric:

$$\frac{\partial}{\partial t} g_{ij} = -2\mathrm{Ric}_{ij} + \rho g_{ij}.$$

For 2D surfaces, $\mathrm{Ric}_{ij} = K g_{ij}$, where K is the Gaussian curvature. The scalar ρ dictates the type of flow:

- $\rho = 0$ **(Unnormalized Ricci Flow):** The space shrinks or expands depending on the sign of curvature. Positive curvature shrinks to a point.
- $\rho = \rho_t$ **(Normalized Ricci Flow):** Here, ρ_t is the time-dependent average curvature. This formulation preserves the total area of the surface as it evolves.
- $\rho =$ **constant (Yamabe Flow):** Evolving the metric by a constant scalar times the metric.

Ricci Flow on Surfaces

For our 2D conformal metric $g = u(x, y, t)(dx^2 + dy^2)$, the unnormalized Ricci flow equation simplifies drastically.
Since $\frac{\partial g}{\partial t} = -2Kg$, and $g = u(dx^2 + dy^2)$, the weight function evolves according to:

$$\frac{\partial u}{\partial t} = -2Ku.$$

Substitute $K = -\frac{1}{2u}\Delta(\ln u)$, and we get a diffusion-like equation for the conformal factor!